Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Information Technology

Continuous Assessment Test – II

Question Paper

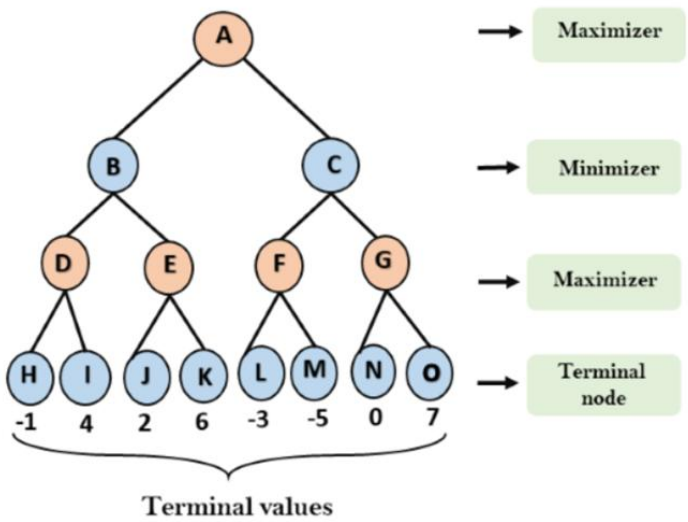
Degree & Branch	B.Tech. Information Technology	**Semester**	V
Subject Code & Name	**UIT2504 Artificial Intelligence**		
Time: 90 Minutes **Date: 7-11-2023**	**Answer All Questions**	**Maximum: 50 Marks**	

Course Outcomes:

CO1:Understand the basic concepts of intelligent agents
CO2: Develop general-purpose problem-solving agents, logical reasoning agents, and agents that reason under uncertainty
CO3: Employ AI techniques to solve some of today's real-world problems

Part – A (4 × 2 = 8 Marks)

KL- Knowledge Level CO – course Outcome PI – Performance Indicator

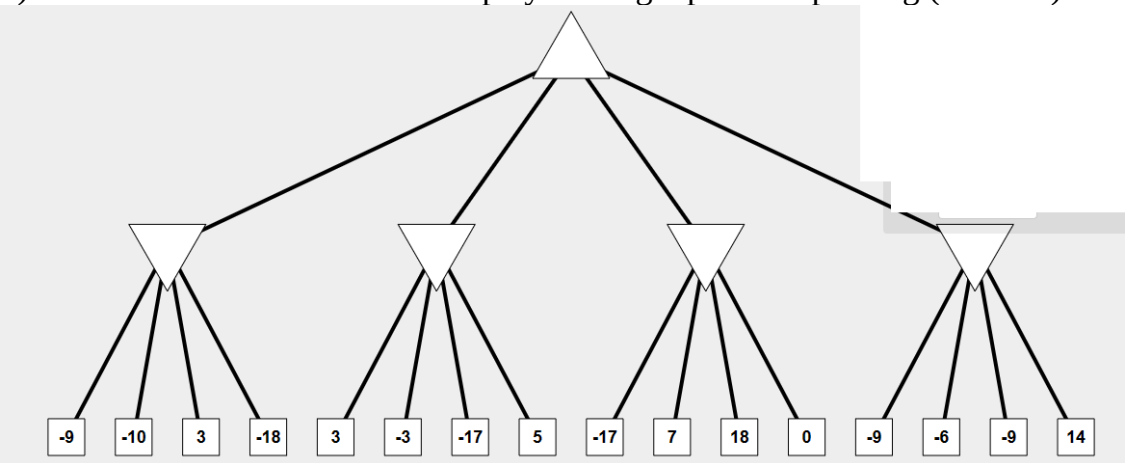
S no	Questions	KL	CO	PI
1	Apply the minimax algorithm for the following and find the next move for the max player  	2	1	1.1.2
2	Justify why it is a good heuristic to choose the variable that is most constrained but the value that is least constraining in a CSP search.	2	1	1.1.2
3	Which of the following propositions are tautologies? Which are contradictions? Why? • Three is a prime number, • It is raining or it is not raining, • It is raining (P) and it is not raining (¬P)	2	2	2.1.3
4	Translate the following Propositional Logic to English sentences where :	2	2	2.1.3

	Proposition p: Alice is smart, Proposition q: Alice is honest. • $\neg p \wedge q$ • $p \vee (\neg p \wedge q)$ • $p \rightarrow \neg q$			

Part – B (3 × 6 = 18 Marks)

5	How is online search agent different from offline search agent. Explain in detail the working of online DFS agent with suitable example	2	1	1.1.2 2.1.3
6	We have five planes: A, B, C, D, and E and two runways: international and domestic. We would like to schedule a time slot and runway for each aircraft to either land or take off. We have four time slots: 1, 2, 3, 4 for each runway, during which we can schedule a landing or take off of a plane. We must find an assignment that meets the following constraints: • Plane B has lost an engine and must land in time slot 1. • Plane D can only arrive at the airport to land during or after time slot 3. • Plane A is running low on fuel but can last until at most time slot 2. • Plane D must land before plane C takes off, because some passengers must transfer from D to C. • No two aircrafts can reserve the same time slot for the same runway. Complete the formulation of this problem as a CSP in terms of variables, domains, and constraints (both unary and binary). Constraints should be expressed implicitly using mathematical or logical notation rather than with words. Make sure to specify variables, domains, and constraints.	3	3	1.1.2 2.1.3 14.1.1
7	When are Monte Carlo Tree search preferred? Explain the various steps in Monte Carlo tree search with suitable real time example	2	2	1.1.2 2.1.3

Part – C (2 × 12 = 24 Marks)

8	a) What are the drawbacks of Minimax Search technique? How are the drawbacks handled by Alpha Beta Pruning? Explain the working of Alpha beta pruning with appropriate algorithm (5 marks) b) Find the next move of the max player using alpha beta pruning (7 marks)  	3	1	1.1.2 2.1.3
OR				

9	a) Solve the map coloring problem as constraint satisfaction problem, where each of the country is to be colored with either red, green or blue. Illustrate the step by step expansion using backtracking algorithm (5 marks) b) Explain with suitable algorithm how Arc3 consistency check can improve backtracking in CSP (7 marks)	3	1	1.1.2 2.1.3
10	Consider the following English statements: • If Alice goes to the store or Bob goes to the park, then Carol will visit the museum. • If Carol visits the museum, then David will go to the zoo or Emily will go to the theater. • If Emily goes to the theater, then Frank will go to the concert. • If David goes to the zoo, then Carol will not visit the museum. • Carol is not visiting the museum. • Frank is not going to the concert. • If Alice goes to the store, then Emily will go to the theater. • David is going to the zoo. • Alice is going to the store. Using propositional logic resolution, determine whether the statement "Frank is going to the concert" is valid based on the given statements.	3	2	1.1.2 2.1.3 14.1.1
	OR			
11	Imagine a complex scenario involving a murder investigation with the following English statements: • If there is a motive and opportunity, then someone committed the murder. • If the suspect has no alibi, then there is an opportunity. • If the victim and suspect had a conflict, then there is a motive. • If there is physical evidence, the case is solved. • There is no alibi for the suspect. • The victim and suspect had a conflict. • There is physical evidence. • The case is not solved. Use propositional logic to perform both forward chaining and backward chaining to determine the validity of the statement "Someone committed the murder."	3	2	1.1.2 2.1.3 14.1.1

Prepared By Course Instructor	Reviewed By Panel Members	Verified by Program Coordinator	Approved By HoD, IT
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